Soft-MoE LoRA with Hybrid Modulation for Night-Robust Multi-Modal Maritime Segmentation

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Abstract

This paper describes our solution for the MaCVi USV Semantic Segmentation Challenge at CVPR 2026, which targets the MULTIAQUA dataset featuring RGB, thermal, and LiDAR modalities with a challenging nighttime test set. We adapt the MemorySAM framework—which repurposes SAM2's temporal memory mechanism for cross-modal feature aggregation—to the maritime domain and investigate effective strategies for multi-modal fusion under severe illumination changes. Specifically, we adopt soft mixture-of-experts gating in SAM2's LoRA layers to improve expert utilization when the number of experts is small, and introduce a cross-modal fusion head that jointly compares all modality features to produce relative importance weights, replacing independent per-modality scoring that suffers from sigmoid saturation. The resulting weights are used with max-normalization for memory modulation—preserving the best modality's features while suppressing degraded ones—and as raw softmax weights for output fusion. To bridge the day-night domain gap without nighttime training data, we apply aggressive RGB corruption augmentations including extreme darkening, complementary random masking [11], and stochastic full-channel zeroing. We empirically validate each design choice and report our results on the MULTIAQUA benchmark.

1. Introduction

Reliable scene understanding is a critical capability for unmanned surface vehicles (USVs) operating in maritime environments. Unlike structured road scenes, waterways present unique challenges: water surfaces exhibit highly variable reflectance, obstacles appear at unpredictable scales and distances, and visibility conditions can shift dramatically between day and night. Multi-modal sensing—combining RGB cameras with thermal and LiDAR sensors—offers a principled approach to address these challenges, as each modality provides complementary information robust to different environmental degradations [14, 15].

However, effectively fusing heterogeneous modalities remains an open problem. Traditional multi-modal segmentation methods [5, 9, 15] design modality-specific encoders or fusion modules, which scale poorly as the number of modalities increases. Meanwhile, foundation models such as SAM [4] and SAM2 [8] have demonstrated remarkable segmentation capabilities through large-scale pretraining, but they are fundamentally designed for single-modal (RGB) input and instance-level tasks. Bridging foundation models to multi-modal semantic segmentation thus requires addressing two fundamental gaps: enabling multi-modal input processing and transitioning from instance-level to semantic-level understanding.

Recently, MemorySAM [1] proposed an elegant solution by reinterpreting SAM2's temporal memory mechanism for cross-modal aggregation. By treating each modality as a sequential "frame", the memory bank naturally captures modality-agnostic features, while a Semantic Prototype Memory Module (SPMM) guides the model toward semantic-level segmentation. Concurrently, MoE-LoRA [16] introduced Mixture-of-Experts routing into SAM's Low-Rank Adaptation [3] layers, enabling modality-adaptive parameter specialization with minimal overhead.

In this work, we build upon these advances and present our solution for the **4th Workshop on Maritime Computer Vision (MaCVi)** USV Semantic Segmentation Challenge at CVPR 2026. The challenge employs the MULTIAQUA dataset [6], which provides synchronized RGB, thermal, and LiDAR data captured from USVs across daytime and nighttime conditions, with four semantic classes: static obstacle, dynamic obstacle, water, and sky. A key difficulty is that models are trained on daytime data but must generalize to a nighttime test set where RGB sensors become severely degraded.

Our approach adapts the MemorySAM framework to the maritime domain with the following design choices:

(1) Soft-MoE gating for LoRA adaptation. MoE-

LoRA [16] employs hard top-k routing, where only k out of N experts are activated per token. When the number of experts is small (*e.g.*, 3 experts matching 3 modalities), top-k routing discards significant expert capacity. Following insights from Soft Mixtures of Experts [7], we adopt soft gating where all experts contribute via softmax-weighted combination, ensuring full expert utilization in this low-expert regime.

(2) Cross-modal scoring with dual-path modulation. Prior approaches [1] estimate each modality's quality independently, which can suffer from sigmoid saturation—when all logits are large, scores converge to $\sim$1.0 regardless of actual quality differences. We replace independent scoring with a cross-modal fusion head that jointly compares all modality features via a shared compression and comparison network, producing softmax-normalized relative weights. These weights are then applied through two distinct paths: max-normalized for memory modulation (preserving the best modality at full strength) and as raw softmax for output fusion (weighting contributions proportionally).

(3) Night-robust RGB augmentation. To bridge the day-night domain gap using only daytime training data, we apply aggressive RGB corruption including extreme brightness reduction (factor 0.03–0.25), complementary random masking [11] that zeros out 20–50% of spatial regions, and stochastic full-channel zeroing simulating complete RGB failure. These augmentations encourage the model to rely on thermal and LiDAR when RGB is degraded.

We empirically validate each component through controlled experiments and report our challenge results on the MULTIAQUA benchmark. The remainder of this paper is organized as follows: Sec. 2 reviews related work, Sec. 3 details our method, and Sec. 4 presents experimental results.

2. Related Work

2.1. Multi-Modal Semantic Segmentation

Multi-modal semantic segmentation aims to leverage complementary information from heterogeneous sensors to improve scene understanding beyond what any single modality can achieve. Early approaches [2, 12] adopted dual-encoder architectures with element-wise or concatenation-based fusion, but these methods are typically designed for fixed modal pairs (*e.g.*, RGB-Depth or RGB-Thermal) and cannot easily generalize to arbitrary combinations.

Transformer-based architectures have significantly advanced this field. SegFormer [13] introduced a hierarchical Transformer encoder with a lightweight MLP decoder, eliminating positional encoding to enable flexible multi-scale feature extraction. Building on this backbone, CMX [14] proposed a unified cross-modal fusion framework with a Cross-Modal Feature Rectification Module (CM-FRM) and a Feature Fusion Module (FFM) that performs long-range context exchange across modalities, achieving state-of-the-art results across RGB-Depth, RGB-Thermal, RGB-Polarization, and RGB-LiDAR benchmarks. CMNeXt [15] further scaled this paradigm with the Self-Query Hub (SQ-Hub) and Parallel Pooling Mixer (PPX), enabling fusion across an arbitrary number of modalities with negligible per-modality parameter overhead and demonstrating +9.10% mIoU improvement with quad-modal input on the DELIVER benchmark.

StitchFusion [5] took a different approach by integrating large-scale pretrained models as both encoders and feature fusers, using a MultiAdapter module for cross-modal information transfer during encoding. This enables multi-modal fusion with minimal additional parameters by leveraging pretrained representations directly. MMSFormer [9] introduced a novel fusion block within a Transformer architecture that progressively improves performance as additional modalities are incorporated, demonstrating the effectiveness of modality-agnostic fusion on material and semantic segmentation benchmarks.

For modality robustness, Shin *et al*. [11] proposed Complementary Random Masking (CRM) for RGB-Thermal segmentation, which strategically masks input modalities during training to prevent over-reliance on any single sensor. Combined with a self-distillation loss, CRM forces the network to learn complementary representations, improving robustness when individual modalities are degraded. Our night augmentation pipeline draws inspiration from this approach, extending it with extreme brightness corruption and full-channel zeroing tailored for maritime nighttime scenarios.

For maritime environments specifically, the MULTIAQUA dataset [6] provides synchronized RGB, thermal, and LiDAR data across daytime and nighttime conditions. It introduces robust training strategies that enable networks trained solely on daytime images to generalize to nighttime scenarios, highlighting the importance of augmentation-based domain adaptation for multi-modal maritime perception.

2.2. Foundation Model Adaptation for Segmentation

The Segment Anything Model (SAM) [4] established a new paradigm for promptable image segmentation through large-scale pretraining on over 1 billion masks. Its successor, SAM2 [8], extended this capability to video with a streaming memory architecture that maintains temporal context through memory banks and cross-attention mechanisms. However, both models are designed for single-modal (RGB) input and focus on instance-level segmentation, requiring adaptation for multi-modal semantic tasks.

Low-Rank Adaptation (LoRA) [3] provides a parameter-efficient strategy for adapting large pretrained models by injecting trainable low-rank matrices into frozen attention layers. Applied to SAM, LoRA enables task-specific adaptation while preserving the foundation model's generalization capability with minimal additional parameters.

MoE-LoRA [16] advanced this approach by incorporating Mixture-of-Experts (MoE) [10] routing into the LoRA framework for multi-modal SAM adaptation. By maintaining multiple expert pairs and using a gating network for top-k selection, it achieves modality-adaptive feature extraction with a 32.15% improvement under missing modality conditions. However, the hard top-k routing can lead to underutilized experts when the number of experts is small, which is common when the expert count matches the number of input modalities. Recent work on Soft Mixtures of Experts [7] has shown that replacing discrete top-k selection with continuous softmax weighting over all experts improves gradient flow and eliminates dead expert problems, motivating our adoption of soft gating in the LoRA context.

MemorySAM [1] proposed an alternative approach by repurposing SAM2's video memory mechanism for multi-modal fusion. By treating different modalities as sequential frames, the memory bank naturally captures cross-modal features without requiring explicit fusion modules. Combined with a Semantic Prototype Memory Module (SPMM) that maintains category-level prototypes through exponential moving average updates, it bridges the gap from instance-level to semantic-level understanding, achieving state-of-the-art results of 65.38% mIoU on DELIVER and 52.88% on MCubeS.

Our work builds upon MemorySAM's memory-based multi-modal framework while introducing Soft-MoE LoRA for improved expert utilization and a hybrid modulation strategy that separates absolute quality assessment from relative importance ranking.

3. Method

Our method adapts MemorySAM [1] for the MULTIAQUA maritime segmentation task. We retain the core idea of treating each modality as a sequential frame processed through SAM2's [8] memory-based tracking pipeline, and modify the LoRA adaptation strategy, modality scoring, and output fusion. Our pipeline operates in four phases: given M input modalities (RGB, thermal, LiDAR), (1) all modalities are first encoded by SAM2's frozen image encoder with Soft-MoE LoRA layers; (2) a cross-modal fusion head jointly compares the backbone features of all modalities and produces softmax-normalized relative weights; (3) these weights are max-normalized and used to modulate vision features before SAM2's memory-augmented tracking step (UAMM); and (4) the per-modality mask outputs are fused using the raw softmax weights (AMF).

3.1. Soft-MoE LoRA

We keep SAM2's image encoder frozen and inject low-rank adaptation matrices into the query and value projections of each attention block, following the standard LoRA [3] formulation. Rather than using a single pair of low-rank matrices, we maintain N expert pairs $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^N$, where $\mathbf{A}_i \in \mathbb{R}^{d\times r}$ and $\mathbf{B}_i \in \mathbb{R}^{r\times d}$, with r being the LoRA rank and d the feature dimension.

Following the Soft Mixture-of-Experts formulation [7], instead of selecting the top-k experts via hard routing [10, 16], we compute a softmax-weighted combination over all experts for each input token $\mathbf{x}$:

$$\Delta\mathbf{x} = \sum_{i=1}^{N} \alpha_i(\mathbf{x}) \cdot \mathbf{B}_i \mathbf{A}_i \mathbf{x}, \quad \alpha_i(\mathbf{x}) = \frac{\exp(g_i(\mathbf{x}))}{\sum_{j=1}^{N} \exp(g_j(\mathbf{x}))}, \tag{1}$$

where $g(\mathbf{x}) = \mathbf{W}_g\mathbf{x} + \mathbf{b}_g$ is a lightweight gating network with $\mathbf{W}_g \in \mathbb{R}^{N\times d}$. The output of the original QKV projection is augmented as $\mathbf{q} \leftarrow \mathbf{q} + \Delta\mathbf{q}$ and $\mathbf{v} \leftarrow \mathbf{v} + \Delta\mathbf{v}$.

When the number of experts N is small (*e.g.*, N=3 matching three modalities), hard top-k routing with k=2 permanently disables one expert per token. Soft gating avoids this issue by ensuring gradient flow to every expert at every step, enabling all experts to specialize without the dead expert problem. We use N=3 experts and r=4 throughout our experiments.

3.2. Cross-Modal Scoring

A central challenge in multi-modal fusion is estimating the relative quality of each modality. Prior approaches [1] use independent per-modality heads (*e.g.*, a CNN followed by sigmoid activation) to produce scalar confidence scores. However, when the backbone features are rich and all modalities produce large logits, the sigmoid function saturates—all scores converge to $\sim$1.0 regardless of actual quality differences. This effectively disables modality-adaptive weighting.

We replace the independent scoring mechanism with a cross-modal fusion head that jointly observes all modality features. Given backbone feature maps $\{\mathbf{F}_i\}_{i=1}^M$ from Phase 1, a shared compression network ϕ maps each to a compact vector via global average pooling and a linear projection:

$$\mathbf{z}_i = \phi(\mathbf{F}_i) = \mathrm{ReLU}(\mathbf{W}_c \cdot \mathrm{GAP}(\mathbf{F}_i) + \mathbf{b}_c), \tag{2}$$

where $\mathbf{W}_c \in \mathbb{R}^{h\times C}$ is shared across modalities with hidden dimension h. The compressed vectors are concatenated and passed through a comparison MLP to produce modality logits:

$$\mathbf{l} = \psi([\mathbf{z}_1; \mathbf{z}_2; \ldots; \mathbf{z}_M]), \quad w_i = \frac{\exp(l_i/\tau)}{\sum_{j=1}^{M} \exp(l_j/\tau)}, \tag{3}$$

where ψ is a two-layer MLP and τ is a temperature parameter. The softmax normalization inherently produces relative weights: even when all modalities are high quality, the network can express fine-grained preferences; when one modality degrades, its weight decreases and others increase. We initialize the output layer of ψ to zero, so that $\mathbf{l} = \mathbf{0}$ at the start of training and all modalities receive uniform weight $1/M$, ensuring stable initialization.

3.3. Uncertainty-Aware Memory Modulation

Before each modality's features enter SAM2's memory-augmented tracking step, we modulate them based on the cross-modal weights. To preserve the best modality's features at full strength while suppressing degraded ones, we apply max-normalization to the softmax weights:

$$\hat{w}_i = \frac{w_i}{\max_j w_j + \epsilon}, \quad \hat{\mathbf{f}}_i^{(k)} = \hat{w}_i \cdot \mathbf{f}_i^{(k)}, \quad \forall k, \tag{4}$$

where $\mathbf{f}_i^{(k)}$ are the multi-scale vision features for modality i and $\epsilon = 10^{-8}$.

The max-normalization ensures that the highest-scoring modality always receives $\hat{w} = 1.0$ (no attenuation), while other modalities are scaled down proportionally. This addresses the sigmoid saturation problem: unlike sigmoid-based gating where all scores can converge to $\sim$1.0, the softmax-based relative comparison always differentiates between modalities, and max-normalization preserves the absolute feature magnitude of the best modality. For instance, if thermal is the most reliable modality at night ($w_{\text{thermal}} = 0.6$, $w_{\text{lidar}} = 0.3$, $w_{\text{rgb}} = 0.1$), the UAMM scores become $\hat{w}_{\text{thermal}} = 1.0$, $\hat{w}_{\text{lidar}} = 0.5$, $\hat{w}_{\text{rgb}} = 0.17$—thermal features pass through unmodified while RGB is strongly suppressed.

The modulated features $\hat{\mathbf{f}}_i$ are then processed by SAM2's `track_step`, which updates the memory bank with cross-attention. The first modality serves as the initialization frame, and subsequent modalities attend to accumulated memory, enabling cross-modal feature exchange through SAM2's built-in temporal attention mechanism.

3.4. Adaptive Mask Fusion

After obtaining per-modality mask predictions $\mathbf{o}_i$ from the tracking loop, we fuse them using the raw softmax weights from the cross-modal fusion head:

$$\mathbf{o}_{\text{fused}} = \sum_{i=1}^{M} w_i \cdot \mathbf{o}_i, \tag{5}$$

where w_i are the softmax weights from Eq. (3), which already sum to one.

This dual-path design—max-normalized weights for UAMM, raw softmax weights for AMF—serves complementary purposes. UAMM requires *absolute* preservation: the best modality should contribute its full feature magnitude to the memory bank. AMF requires *relative* weighting: each modality's mask output should be weighted proportionally to its estimated quality. Reusing the same cross-modal fusion head for both paths avoids redundant computation while tailoring the weight application to each stage's requirements.

3.5. Prototype Segmentation Loss

Following MemorySAM [1], we use a Semantic Prototype Memory Module to guide SAM2 toward semantic-level understanding. For each class c, a global prototype $\mathbf{p}_c$ is maintained via exponential moving average (momentum $= 0.8$). At each training step, batch prototypes $\hat{\mathbf{p}}_c$ are computed by averaging the fused feature vectors $\mathbf{F}_{\text{fused}}$ belonging to class c (determined by resizing the ground-truth labels to the feature map resolution). The prototype loss is defined as:

$$\mathcal{L}_{\text{proto}} = \text{MSE}(\hat{\mathbf{p}}, \mathbf{p}), \tag{6}$$

and the total training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{OHEM}} + \lambda \cdot \mathcal{L}_{\text{proto}}, \tag{7}$$

where $\mathcal{L}_{\text{OHEM}}$ is the Online Hard Example Mining cross-entropy loss and λ is a scaling factor.

3.6. Night-Robust RGB Augmentation

The MULTIAQUA dataset requires generalization from daytime training to nighttime evaluation, where RGB quality degrades severely while thermal and LiDAR remain relatively stable. Inspired by the augmentation strategies proposed in the MULTIAQUA paper [6] and CRM [11], we apply three complementary RGB corruptions during training (applied only to the RGB modality, leaving thermal and LiDAR unchanged):

Night Simulation (probability 0.35): extreme brightness reduction (factor sampled from $[0.03, 0.25]$), followed by contrast reduction ($[0.3, 0.7]$), gamma correction ($\gamma \in [0.4, 0.8]$), and additive Gaussian noise ($\sigma = 0.02$).

Complementary Random Masking (probability 0.30): rectangular patches are randomly placed on the RGB image, zeroing out 20–50% of spatial regions. This forces the model to rely on thermal and LiDAR for the masked areas.

Full-Channel Zeroing (probability 0.12): the entire RGB input is replaced with zeros, simulating complete RGB failure. This is the most extreme case, requiring the model to produce reasonable segmentation from thermal and LiDAR alone.

These augmentations are stacked sequentially: an image may undergo night simulation and then additionally be partially masked, creating diverse degradation patterns.

Table 1. Main results on the MULTIAQUA benchmark. Daytime val mIoU and nighttime test mIoU are reported. †Results from the MaCVi challenge leaderboard.

Method	Modalities	Val mIoU	Test mIoU
CMNeXt [15]	RGB+T+L	**[TODO: XX.X]**	**[TODO: XX.X]**
Ours	RGB+T+L	94.00	**[TODO: XX.X]**

Table 2. Per-class IoU (%) on the MULTIAQUA validation set.

	Static	Dynamic	Water	Sky	mIoU
Ours	97.25	80.82	99.08	98.84	94.00

Table 3. Ablation on night-robust RGB augmentation. "Night Aug" includes night simulation, CRM, and full-channel zeroing.

Night Aug	Val mIoU	Test mIoU
✗	**[TODO: XX.X]**	**[TODO: XX.X]**
✓	94.00	**[TODO: XX.X]**

Table 4. Ablation on input modalities.

Modalities	Val mIoU	Test mIoU
RGB only	**[TODO: XX.X]**	**[TODO: XX.X]**
RGB + Thermal	**[TODO: XX.X]**	**[TODO: XX.X]**
RGB + Thermal + LiDAR	94.00	**[TODO: XX.X]**

4. Experiments

4.1. Dataset and Setup

Dataset. We evaluate on the MULTIAQUA dataset [6], which provides synchronized RGB, thermal, and LiDAR data captured from unmanned surface vehicles. The dataset contains four semantic classes: static obstacle, dynamic obstacle, water, and sky. Models are trained on daytime data and evaluated on both a daytime validation set and a nighttime test set, where RGB quality degrades significantly.

Implementation details. We use SAM2 (Hiera-based backbone) as the foundation model with LoRA rank $r{=}4$ and $N{=}3$ Soft-MoE experts. All images are resized to 1024×1024. We train for 200 epochs with AdamW optimizer (learning rate 6×10^{-4}, weight decay 0.01) using a warmup-polynomial learning rate schedule (10 warmup epochs, power 0.9). The batch size is 1 per GPU with gradient accumulation, and we use Online Hard Example Mining (OHEM) cross-entropy loss with the prototype segmentation loss (Sec. 3.5). SAM2's image encoder is frozen; only the Soft-MoE LoRA layers, cross-modal fusion head, prompt encoder, and mask decoder are trained. Training is conducted with PyTorch DDP on NVIDIA GPUs.

Evaluation metric. We report mean Intersection-over-Union (mIoU) averaged over the four semantic classes.

4.2. Main Results

Tab. 1 presents our results on the MULTIAQUA benchmark. On the daytime validation set, our method achieves 94.00% mIoU.

Tab. 2 shows the per-class breakdown on the validation set. The model achieves strong performance on water (99.08%) and sky (98.84%), which occupy large contiguous regions, and on static obstacles (97.25%). The dynamic obstacle class is the most challenging at 80.82%, which is expected given the smaller object sizes and higher appearance variability of moving vessels.

4.3. Ablation Studies

Effect of night augmentation. To validate the importance of our night-robust augmentation pipeline (Sec. 3.6), we train an identical model without any RGB corruption augmentations and compare performance.

Effect of modality combination. We investigate the contribution of each modality by training models with different subsets.

5. Conclusion

We presented our solution for the MaCVi USV Semantic Segmentation Challenge at CVPR 2026. By adapting MemorySAM's memory-based multi-modal framework to the maritime domain, we investigated soft mixture-of-experts gating for LoRA adaptation, a cross-modal fusion head that jointly compares all modality features to produce relative importance weights with dual-path application—max-normalized for memory modulation and raw softmax for output fusion—and aggressive RGB corruption augmentations for day-to-night generalization. Our experiments on the MULTIAQUA benchmark validate the effectiveness of these design choices for robust multi-modal maritime segmentation.

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